WORKED SOLUTION

STEP II 2018, Question 2

Pure • Concave Functions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q2

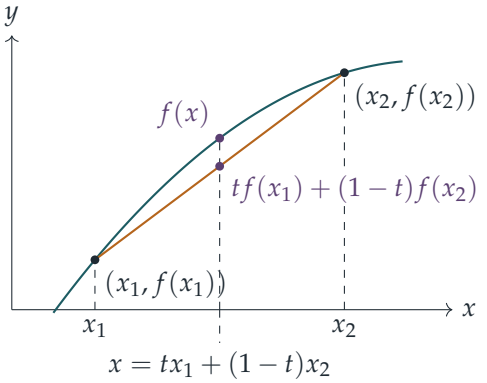
The key idea

The point $tx_1 + (1-t)x_2$ runs from x_2 to x_1 as t goes from 0 to 1, and $tf(x_1) + (1-t)f(x_2)$ is the height of the chord above that point. So concavity says that every chord lies on or below the curve. Once that two-point inequality is in hand, a three-point version comes from using it twice, and both triangle inequalities follow by applying it to a suitable concave function with $A + B + C = \pi$.

Solution

The sketch

Write $x = tx_1 + (1-t)x_2$. Then the point $(x,\ tf(x_1) + (1-t)f(x_2))$ divides the chord from $(x_2, f(x_2))$ to $(x_1, f(x_1))$ in the ratio $(1-t) : t$, so it lies on the chord directly above (or below) x. The definition says the curve at x is at least as high as the chord.



Why $f'' < 0$ gives concavity

If $x_1 = x_2$, or $t = 0$ or $t = 1$, the inequality is an equality, so take $0 < t < 1$ and, without loss of generality, $x_1 < x_2$ (swapping x_1, x_2 just swaps t and $1 - t$). Then $x = tx_1 + (1-t)x_2$ satisfies $x_1 < x < x_2$, with

$$x - x_1 = (1-t)(x_2 - x_1), \qquad x_2 - x = t(x_2 - x_1).$$

By the mean value theorem there are $c_1 \in (x_1, x)$ and $c_2 \in (x, x_2)$ with

$$\frac{f(x) - f(x_1)}{x - x_1} = f'(c_1), \qquad \frac{f(x_2) - f(x)}{x_2 - x} = f'(c_2).$$

Since $f'' < 0$ on (a, b), f' is strictly decreasing there, and $c_1 < c_2$, so $f'(c_1) > f'(c_2)$. Substituting the lengths above and cancelling the positive factor $x_2 - x_1$,

$$\frac{f(x) - f(x_1)}{1-t} > \frac{f(x_2) - f(x)}{t}.$$

Multiplying by $t(1-t) > 0$ gives $tf(x) - tf(x_1) > (1-t)f(x_2) - (1-t)f(x)$, that is,

$$f(x) > tf(x_1) + (1-t)f(x_2).$$

So f is concave on (a, b). In words: the gradient of the curve keeps falling, so the slope from x_1 to x is bigger than the slope from x to x_2, and the curve bulges above the chord.

Part (i)

? **Ask yourself** Which two points, with which weight, average to $\frac{1}{3}(u+v+w)$?

Group v and w together: take $x_1 = u$, $x_2 = \frac{1}{2}(v+w)$ and $t = \frac{1}{3}$. Then x_2 lies in (a, b), since it is between v and w, and

$$tx_1 + (1-t)x_2 = \tfrac{1}{3}u + \tfrac{2}{3} \cdot \tfrac{1}{2}(v+w) = \tfrac{1}{3}(u+v+w).$$

The definition gives

$$f\Big(\frac{u+v+w}{3}\Big) \geq \tfrac{1}{3}f(u) + \tfrac{2}{3}f\Big(\frac{v+w}{2}\Big).$$

Now use the definition again with $x_1 = v$, $x_2 = w$, $t = \frac{1}{2}$: $f(\frac{1}{2}(v+w)) \geq \frac{1}{2}f(v) + \frac{1}{2}f(w)$. Since this term has the positive coefficient $\frac{2}{3}$, substituting keeps the inequality the same way round:

$$f\Big(\frac{u+v+w}{3}\Big) \geq \tfrac{1}{3}f(u) + \tfrac{2}{3} \cdot \tfrac{1}{2}(f(v) + f(w)) = \frac{f(u) + f(v) + f(w)}{3},$$

which is the required result.

Part (ii)

Take $f(x) = \sin x$ on $0 < x < \pi$. Here $f''(x) = -\sin x < 0$, so sin is concave on $(0, \pi)$. The angles of a triangle satisfy $0 < A, B, C < \pi$ and $A + B + C = \pi$, so part (i) with $u = A$, $v = B$, $w = C$ gives

$$\frac{\sin A + \sin B + \sin C}{3} \leq \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2},$$

so $\sin A + \sin B + \sin C \leq \dfrac{3\sqrt{3}}{2}$.

Part (iii)

For $0 < x < \pi$ we have $\sin x > 0$, so $g(x) = \ln(\sin x)$ is defined there, with

$$g'(x) = \frac{\cos x}{\sin x} = \cot x, \qquad g''(x) = -\operatorname{cosec}^2 x < 0.$$

So g is concave on $(0, \pi)$, and part (i) gives

$$\frac{\ln \sin A + \ln \sin B + \ln \sin C}{3} \leq \ln \sin \frac{\pi}{3} = \ln \frac{\sqrt{3}}{2}.$$

Multiplying by 3 and combining the logarithms, $\ln(\sin A \sin B \sin C) \leq \ln\left(\frac{\sqrt{3}}{2}\right)^3$. Since exp is increasing,

$$\sin A \times \sin B \times \sin C \leq \left(\frac{\sqrt{3}}{2}\right)^3 = \frac{3\sqrt{3}}{8}.$$

Checks

Both bounds are attained by the equilateral triangle: $3 \sin \frac{\pi}{3} = \frac{3\sqrt{3}}{2} \approx 2.598$ and $\sin^3 \frac{\pi}{3} = \frac{3\sqrt{3}}{8} \approx 0.6495$. A right-angled isosceles triangle gives a sum of $1 + \sqrt{2} \approx 2.414$ and a product of $\frac{1}{2}$, both below the bounds. A computer check over 200 000 random triangles found no sum above 2.598076 and no product above 0.649519. A very flat triangle ($A, B \to 0$, $C \to \pi$) sends both towards 0, as it should.

Common slips

- In part (i), using $t = \frac{1}{3}$ with $x_1 = u$ and $x_2 = v$ gives $\frac{1}{3}u + \frac{2}{3}v$, not the mean of three numbers; the second point must be the midpoint of v and w.
- In part (iii), you can't apply part (ii)'s inequality to the product directly; it is the logarithm that turns the product into a sum. Check also that $\ln \sin x$ is defined, which needs $\sin x > 0$.
- Saying that $f'' < 0$ "means the curve bends down" is not by itself a proof of the chord inequality; the mean value theorem argument makes it precise.

Beyond STEP

Part (i) is the three-point case of Jensen's inequality: for concave f and weights $t_i \geq 0$ summing to 1, $f\left(\sum t_i x_i\right) \geq \sum t_i f(x_i)$, proved by induction exactly as here. With $f = \ln$ it gives the AM–GM inequality, and in probability it says $f(\mathrm{E}[X]) \geq \mathrm{E}[f(X)]$ for concave f. Convex and concave functions, and the mean value theorem behind the second paragraph, are treated carefully in Bartle and Sherbert, *Introduction to Real Analysis*.

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